Spatial and Semantic Consistency Contrastive Learning for Self-Supervised Semantic Segmentation of Remote Sensing Images

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***Abstract*— A critical requirement for the success of supervised deep learning lies in having numerous annotated images, which is often challenging to fulfill in remote sensing semantic segmentation tasks. Self-supervised contrastive learning (CL) offers a strategy for learning general feature representations by pretraining neural networks on vast amounts of unlabeled data and subsequently fine-tuning them on downstream tasks with limited annotations. However, the vast majority of CL methods are designed based on instance discriminative pretext tasks, focusing solely on learning the global representation of the entire image while disregarding the essential spatial and semantic correlations crucial for semantic segmentation tasks. To address the above issues, in this article, we propose a spatial and semantic consistency CL (SSCCL) framework for the semantic segmentation task of remote sensing images. Specifically, a consistency branch in SSCCL is designed to learn feature representations with spatial and semantic consistency by maximizing the similarity of the overlapping regions of the two augmented views. In addition, an instance branch is introduced to learn global representations by enforcing the similarity of two augmented views from one image. Through the integration of the consistency branch and instance branch, the proposed SSCCL framework can learn robust and informative feature representations for semantic segmentation in remote sensing scenarios. The proposed method was evaluated on three publicly available remote sensing semantic segmentation datasets, and the experiment results show that our method achieves superior segmentation performance with limited annotations compared to state-of-the-art CL methods as well as the ImageNet pretraining method.**

***Index Terms*— Contrastive learning (CL), remote sensing images, self-supervised, semantic segmentation.**

I. Introduction

SEMANTIC segmentation plays a pivotal role in remote sensing scene understanding, which aims to assign class labels to each pixel in a remote sensing image. This task has widespread applications in various remote sensing scenarios, including urban planning [1], [2], hazard assessment [3], [4], environmental monitoring [5], [6], and precision agriculture [7], [8]. Over the past decade, considerable advancements have been achieved, due to its ubiquitous significance in remote sensing.

Since the introduction of fully convolutional networks (FCNs) [9], deep-learning-based approaches have emerged as the dominant technique for semantic segmentation. However, their exceptional performance comes at the expense of a large number of pixel-level annotations, which are not only costly but also labor-intensive to obtain. Despite the existence of publicly available semantic segmentation datasets, the current volume of training data remains insufficient to address the complexity of large-scale remote sensing feature extraction. Moreover, the majority of these datasets are limited to specific regions, significantly hampering the generalization capabilities of segmentation models. To overcome these challenges, two potential strategies have been explored. The first involves employing semi-supervised training schemes [10] to reduce the reliance on annotated data and facilitate training by generating predicted annotations for unlabeled data. The second strategy entails leveraging transfer learning [11] to transfer knowledge learned from one or more source domains, thereby enhancing performance in the target domain and reducing the need for annotated data. However, both approaches have yielded limited effects and are not readily applicable to practical large-scale scenarios.

Recently, self-supervised contrastive learning (CL) has garnered significant research attention as a method to learn effective visual representations from large sets of unlabeled images. Notably, substantial progress in CL has been made through instance discrimination tasks [12], [13], [14], [15], where a backbone network learns image-level features that remain invariant to specific image augmentations. In this pretext task, instance discrimination aims to maximize the similarity between two augmented views of the same image while minimizing the similarity to views from other images [16], [17], [18]. However, instance discrimination solely focuses on global image-level feature representations, disregarding spatial coherence—a crucial aspect for semantic segmentation tasks that heavily rely on spatial information. Furthermore, most CL methods are tailored for natural scene images, such as those found in the ImageNet dataset [19], which typically consist of a foreground object and a background. In contrast, remote sensing scenarios entail far more complex foreground and background information. As shown in Fig. 1, different augmented views of

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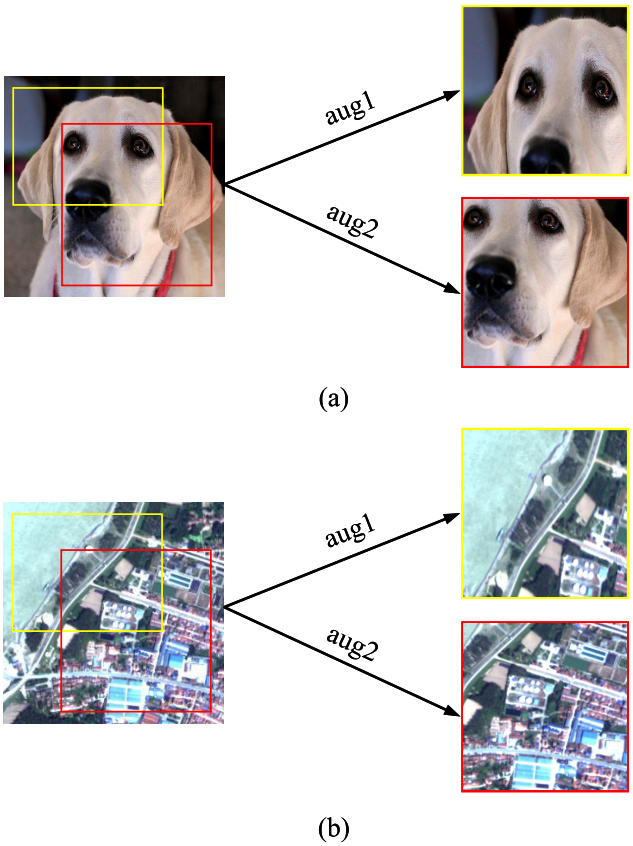


Fig. 1. Randomly cropped and resized views of (a) ImageNet and (b) GID datasets. In (a), the different augmented views depict the same object "dog," while in (b), the different augmented views represent entirely different objects.

a natural image represent the same object, while different augmented views of a remote sensing image have completely different semantic representations. Hence, instance discriminant learning does not apply to remote sensing scenarios, especially for dense prediction tasks such as semantic segmentation.

In this article, we propose a novel CL framework called spatial and semantic consistency CL (SSCCL) specifically tailored for the semantic segmentation task of remote sensing images. To this end, we design a consistency branch within the framework, which aims to learn feature representations exhibiting spatial and semantic consistency. Specifically, the consistency branch maximizes the similarity of overlapping regions between two augmented views. By enforcing consistency within these regions, SSCCL effectively captures the spatial layout and semantic coherence of objects, leading to improved segmentation performance. In addition, to enhance the discriminative power of the learned features, we introduce an instance branch that considers the unique characteristics of individual instances within the image. This branch focuses on capturing instance-level information, which further refines the feature representations and enables better discrimination between different objects and regions. By combining the consistency branch and the instance branch, the proposed SSCCL framework achieves robust and informative feature representations for semantic segmentation in remote sensing scenarios.

The main contributions of this article are summarized as follows.

1) A novel CL framework called SSCCL is proposed in this article, which is specifically designed to tackle the semantic segmentation challenges in remote sensing scenarios. In contrast to existing CL methods, SSCCL addresses the limitations associated with global representations and overlooks spatial and semantic correlations critical for accurate semantic segmentation.
2) Aiming to learn feature representations with both spatial and semantic coherence, we introduce a consistency branch in our proposed SSCCL framework. By maximizing the similarity of overlapping regions between augmented views, the consistency branch effectively captures the intricate spatial layout and semantic coherence of objects within remote sensing images.
3) To further enhance the discriminative power of the learned features, SSCCL introduces an instance branch to refine the feature representations and enable better discrimination between different objects and regions. By combining the consistency branch and the instance branch, the proposed SSCCL framework enables robust and informative feature representations, resulting in superior segmentation performance with limited annotations. We evaluate our proposed method on three publicly available semantic segmentation datasets. Experiment results show that our method outperforms other state-of-the-art CL methods and ImageNet pretraining.

The remainder of the article is organized as follows. Section II presents a discussion of related works on self-supervised learning. In Section III, we provide a detailed description of the proposed SSCCL framework for the self-supervised semantic segmentation task of remote sensing images. Section IV outlines the experiment settings and presents the experiment results. Finally, Section V concludes the article with key findings and implications.

II. Related Work

A. Self-Supervised Learning

Depending on the representation learning paradigm, self-supervised learning can be divided into three main categories as follows: generative, predictive, and contrastive methods.

Generative self-supervised methods learn feature representations by generating or reconstructing input images. Autoencoders [20] were first proposed for the pretraining of artificial neural networks. To learn the multilayer feature maps and filters of the input image, sparse autoencoder [21] and variational autoencoder [22] have since been proposed in succession. Recently, the masked autoencoder (MAE) [23] has made a breakthrough in auto-encoding pretraining of visual transformers by masking random patches of the input image and reconstructing them from potential representations and masked tokens, which has attracted a great deal of attention from the computer vision community.

Predictive self-supervised methods generate labels by designing a specific pretext task and training a model to predict

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these labels and learn the data representations. Relying on the spatial information, Doersch et al. [24] designed the pretext task by predicting the relative positions of randomly cropped patches in one image. Following this paradigm, Noroozi and Favaro [25] built a jigsaw puzzle by disrupting the order of the image patches, and then training the network to predict their correct order. Predicting rotation angle [26] rotated the input image by four independent angles and served as a label for the network prediction, and image inpainting [27] randomly intercepted a patch from the image and allowed the network to learn to recover it.

Contrastive self-supervised methods train the model by comparing two augmented views of the same image and pushing them closer in the representation space. MoCo [28] and SimCLR [17] were two of the first comparison learning methods to learn feature representations by designing an instance discrimination task to find the image with the highest similarity to the input image features in the image set. MoCo designed a momentum encoder module and updated the momentum encoder to abandon the traditional end-to-end training framework, while SimCLR illustrated the importance of a hard positive sampling strategy by introducing data augmentation. BYOL [29] was the first work to use only positive samples for CL, by adding a prediction head to one branch, forming an asymmetric network architecture, and replacing the instance discriminating task with a comparison task. SimSiam [18] also did not need to train the network with negative samples, and the two branches of the network are updated with cross-gradients to avoid network collapse. Besides, Caron et al. [30] first combined the K-means [31] clustering algorithm with self-supervised learning and proposed deep clustering, which generates pseudo-labels by clustering and then uses the pseudo-labels to self-supervise the training of the model. Based on this approach, Caron et al. [16] proposed a cluster-based contrast learning algorithm SwAV, which introduces clustering center information in a synchronous symmetric network architecture to train a feature extraction network.

B. Self-Supervised Learning in Remote Sensing

Recognizing the huge potential of self-supervised learning, there have been numerous attempts to apply it to remote sensing scenarios and to design it based on specific tasks or data sources.

Tile2Vec [32] was proposed for representation learning of remote sensing images with a network trained on triplets of tiles and a triplet loss designed to push neighbor tiles closer in the feature space. Relying on CL, Hou et al. [33] used contrast learning to pretrain hyperspectral image classification models on individual scenes to learn to extract spectral-spatial features, thereby reducing the need for dense annotations. Montanaro et al. [34] performed self-supervised training of an encoder–decoder network to perform land cover classification by learning the feature representation of the encoder through contrast learning and using a perturbation-invariant autoencoder for segmentation training of the decoder. Li et al. [35] designed a local matching contrast loss between patches within images of different views to learn rich pixel-level information for semantic segmentation. SeCo [36] extended the multiaugmented CL approach by taking seasonal factors into account when constructing positive examples and learning rich and transferable remote sensing representations. Chen and Bruzzone [37], [38] use Siamese networks to extract relevant representations from cross-modal remote sensing data and learn pixel feature representations based on shift transformations. Besides, Saha et al. [39] combined deep clustering and triplet CL to learn feature encoding of multitemporal and multimodal remote sensing data for change detection.

III. Proposed Method

In this section, we provide a comprehensive description of the proposed SSCCL framework for self-supervised semantic segmentation of remote sensing images.

In Section III-A, we elaborate on the overall framework and delve into each component of SSCCL. Subsequently, we present the optimization process of the SSCCL framework in Section III-B, highlighting the steps involved in pretraining the model. Finally, in Section III-C, we demonstrate how to perform semantic segmentation with limited annotations based on the model pretrained using the proposed SSCCL framework.

A. SSCCL Framework

Existing state-of-the-art CL frameworks predominantly rely on instance discrimination as a pretext task, where two augmented views originating from the same reference image are encouraged to exhibit similarity in the embedding space through target learning. However, these methods lack the incorporation of spatial and semantic consistency within the learned features, posing a challenge for semantic segmentation tasks that heavily depend on spatial information and semantic correlation [40], [41]. To overcome the above limitations, the proposed SSCCL framework introduces consistency learning alongside the global similarity objective. By doing so, the network becomes capable of acquiring spatially and semantically consistent features in the embedding space, which significantly benefits semantic segmentation tasks.

The overall flowchart of the proposed SSCCL is presented in Fig. 2. For each image within an unlabeled dataset, our SSCCL framework initially generates two augmented views using a series of data augmentations. These two views are then processed by weight-sharing backbone networks to extract essential features. Subsequently, the extracted features from both views are utilized by the instance branch and consistency branch to enforce instance-level and spatial-semantic-level similarity, respectively. Below we provide a detailed description of each component of the SSCCL framework, including augmentation, feature extraction, instance branch, and consistency branch.

1) Augmentation: Given an unlabeled image x from the dataset, two independent data augmentation operators t and t' are randomly sampled from the same family of augmentations $\mathcal{T}$. These operators are then applied to image x to create two correlated views: query x_q and key x_k. The process can be mathematically expressed as

$$x_q = t(x), \quad x_k = t'(x), t, \ t' \sim \mathcal{T}. \tag{1}$$

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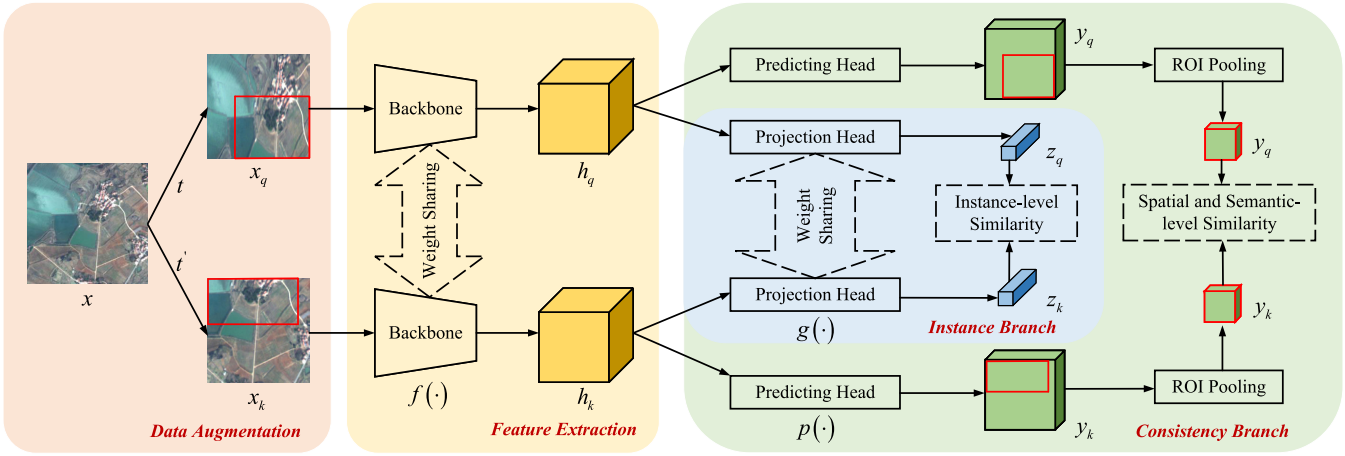


Fig. 2. Overall flowchart of the proposed self-supervised SSCCL framework. Initially, two distinct views are generated by applying data augmentation techniques to the input image. Subsequently, feature representations are extracted from these two views utilizing a backbone network. Finally, the instance branch and consistency branch conduct instance-level and spatial-semantic-level discriminations, respectively. Regarding the instance branch, negative samples are other samples from the mini-batch. As for the consistency branch, all regions, except for the overlapping regions in the two augmented views, are considered negative samples.

Specifically, we generally follow the reference data augmentations in [42], including random horizontal flip, random resized crop, random Gaussian blur, random color jitter, and random grayscale.

Due to the requirement of aligning features from corresponding regions in the consistency branch, random horizontal flips and random resized crops can potentially disrupt the spatial information within the image. Therefore, it is necessary to ensure that both query x_q and key x_k undergo the same angle of flipping and obtain consistent top, left, bottom, and right coordinates (t, l, b, r) for both views.

2) Feature Extraction: After obtaining the two augmented views, a backbone network is employed to extract feature vectors for subsequent CL. Our SSCCL framework does not impose any specific constraints on the selection of the backbone architecture. In this study, we utilize the widely adopted ResNet [43] as the backbone network due to its simplicity and effectiveness.

Formally, let f represent the mapping function of the backbone network. Given the two augmented views x_q and x_k, the features extracted by f can be expressed as follows:

$$h_q = f_q(x_q), \quad h_k = f_k(x_k) \tag{2}$$

where h_q and h_k denote the feature maps extracted from x_q and x_k, respectively, while weights are shared between f_q and f_k.

3) Instance Branch: As shown in Fig. 2, the instance branch comprises a projection head and an instance-level similarity function. A projection head is used to map the extracted features to a high-dimensional representation space. This mapping facilitates better feature separation and discrimination. By projecting features into a higher dimensional space, the instance branch reduces the potential impact of data augmentation on downstream tasks, making the learned representations more robust and transferable to semantic segmentation tasks. Following [17], a multilayer perceptron (MLP) with one hidden layer serves as the projection head, which can be formulated as follows:

$$z_q = g(h_q) = W^{(2)}\sigma\left(W^{(1)}h_q\right), \quad z_k = g(h_k) \tag{3}$$

where $g(\cdot)$ denotes the projection head, z_q and z_k denote the representation vectors, and σ is a ReLU nonlinearity.

In addition, an instance-level similarity optimization function is employed to maximize the similarity between the representation vectors extracted from corresponding augmented views. Specifically, a mini-batch consisting of N samples is augmented to include $2N$ samples. Among these, two different augmented views generated from the same sample are considered positive samples, while the remaining $2(N-1)$ samples serve as negative samples.

To optimize the instance-level similarity between positive samples and minimize the similarity between negative samples, the instance branch selects the InfoNCE [44] loss function as the instance-level similarity loss. This loss function measures the mutual information between positive samples and suppresses the mutual information between negative samples. Instance-level similarity loss encourages the model to learn discriminative and transferable features, as it effectively encourages the network to assign higher similarity scores to positive samples while minimizing the similarity scores of negative samples. The instance-level similarity loss is defined as follows:

$$\mathcal{L}_I = -\log \frac{\ell_i(z_q, z_k)}{\ell_i(z_q, z_k) + \sum_{z_n \in \mathcal{N}_I} \ell_i(z_q, z_n)} \tag{4}$$

with

$$\begin{aligned} \ell_i(z_q, z_k) &= \exp\left(\text{sim}(z_q, z_k)/\tau\right) \\ &= \exp\left(\frac{z_q^{\mathrm{T}} z_k}{\|z_q\|\|z_k\| \cdot \tau}\right) \end{aligned} \tag{5}$$

where $\mathcal{N}_I$ contains vectors of negatives, $\tau > 0$ is a temperature hyper-parameter. $\text{sim}(\cdot)$ represents the cosine similarity function.

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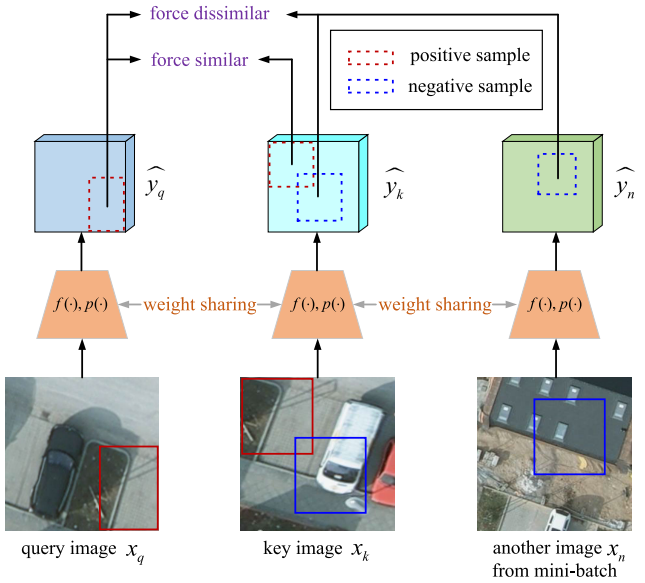


Fig. 3. For the consistency branch, the positive samples are defined as the overlapping regions between the query and key views, while all other regions in the mini-batch are treated as negative samples. The objective of the consistency branch is to maximize the similarity among positive samples and minimize the similarity among negative samples.

4) Consistency Branch: The instance branch focuses solely on the global image-level representation, overlooking the spatial and semantic correlations present in the embedding space. To address this limitation, we introduce a consistency branch that aims to learn feature representations that are both spatially and semantically consistent by aligning the embeddings of a positive pair of views. Specifically, we first employ a prediction head composed of two consecutive 1×1 convolution layers followed by a ReLU nonlinearity to extract dense semantic representation vectors from the feature map

$$y_q = p(h_q) = C^{(2)}\sigma\left(C^{(1)}h_q\right), \quad y_k = p(h_k) \tag{6}$$

where $p(\cdot)$ represents the predicting head and y_q and y_k are dense semantic representation vectors extracted from h_q and h_k.

Subsequently, the overlap region between the two augmented views x_q, x_k on the original image is calculated based on the coordinates (t_q, l_q, b_q, r_q), (t_k, l_k, b_k, r_k) returned from the data augmentation phase. The overlapping rectangular subregions on x_q, x_k are denoted as $\mathcal{R}_q$ and $\mathcal{R}_k$, respectively. As can be seen in Fig. 3, the positive samples for the consistency branch are defined as the regions where the query and key views overlap, while all remaining regions in the mini-batch are considered as negative samples. For the extracted dense semantic representation vectors, we apply Precise RoI Pooling [45] to encode and map them to the same dimension

$$\widehat{y_q} = \phi(y_q, \mathcal{R}_q), \quad \widehat{y_k} = \phi(y_k, \mathcal{R}_k) \tag{7}$$

where ϕ represents the Precise RoI Pooling operation, and $\widehat{y_q}$ and $\widehat{y_k}$ denote the extracted consistency feature vectors of y_q and y_k, which are fixed to the same spatial dimension. With the prediction head and RoI pooling, the consistency branch extracts dense semantic representation vectors from different views, ensuring feature similarity within the same region.

In this article, we develop a spatial-semantic-level similarity function to maximize the pixel-level similarity by pulling the same pixel vectors of $\widehat{y_q}$ and $\widehat{y_k}$ closer. Different from the instance branch, we extend the InfoNCE loss into dense form. The spatial-semantic-level similarity loss can be defined as

$$\mathcal{L}_S = \frac{1}{|\mathcal{P}_S|}\sum_{\widehat{y_q}\in\mathcal{P}_S} -\log\frac{\ell_s\left(\widehat{y_q}, \widehat{y_k}\right)}{\ell_s\left(\widehat{y_q}, \widehat{y_k}\right) + \sum_{\widehat{y_n}\in\mathcal{N}_S}\ell_s\left(\widehat{y_q}, \widehat{y_n}\right)} \tag{8}$$

with

$$\ell_s\left(\widehat{y_q}, \widehat{y_k}\right) = \exp\left(\frac{\widehat{y_q}^{\mathrm{T}}\widehat{y_k}}{\left\|\widehat{y_q}\right\|\left\|\widehat{y_k}\right\| \cdot \tau}\right) \tag{9}$$

where $\mathcal{P}_S$ and $\mathcal{N}_S$ denote pixel vector collections of the positive and negative samples, respectively.

By maximizing the similarity of overlapping regions between augmented views, the consistency branch effectively captures the complex spatial layout and semantic consistency of objects in remote sensing images. Since the remote sensing semantic segmentation task relies on accurate semantic information and spatial location, this spatial and semantic consistency helps to improve the model's ability to distinguish different categories of regions in remote sensing images. Therefore, the consistency branch encourages the network to produce consistent and informative features across different views, thereby enhancing the understanding of spatial structure and semantic relations, which are crucial for accurate semantic segmentation.

B. Self-Supervised Pretraining

The instance branch enhances feature discriminative power by learning object-level distinctions, while the consistency branch captures spatial and semantic consistency within remote sensing images. By combining instance and consensus branches, the synergy between the two helps the model understand fine-grained details and overall scene structure, making SSCCL an efficient framework for remote sensing semantic segmentation with limited annotations. Consequently, the overall loss function of the proposed SSCCL can be defined as follows:

$$\mathcal{L} = \lambda \cdot \mathcal{L}_I + (1-\lambda)\mathcal{L}_S \tag{10}$$

where λ is a hyper-parameter to balance the weight of instance-level similarity loss $\mathcal{L}_I$ and spatial-semantic-level similarity loss $\mathcal{L}_S$. In this article, λ is set to 0.5 by default.

C. Fine-Tune on Semantic Segmentation Task

After performing self-supervised CL training on a large dataset without annotations, the backbone network pretrained by our SSCCL framework acquires the capability to capture high-level representations of the input data. Subsequently, we proceed with fine-tuning the semantic segmentation network, which comprises the pretrained backbone network and a segmentation head, using a limited number of annotated training samples. The overall process is shown in Fig. 4. During fine-tuning, we employ the cross-entropy loss as the objective function for the segmentation task

$$\mathcal{L}_{\text{seg}} = -\frac{1}{H \times W}\sum_{n=1}^{H\times W}\sum_{m=1}^{M} y_{nm}\log(p_{nm}) \tag{11}$$

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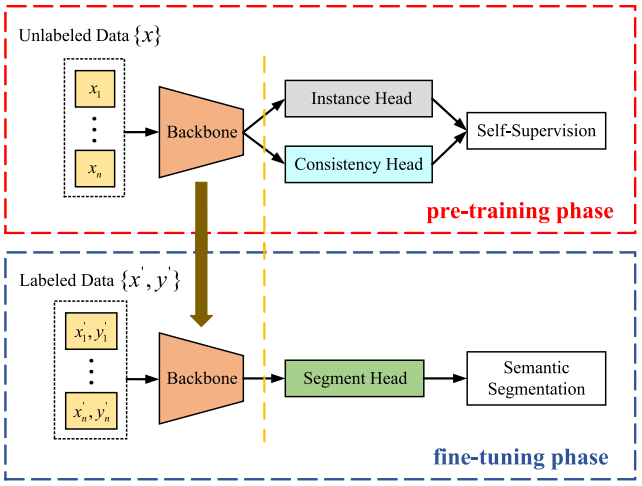


Fig. 4. Pretrained backbone network is integrated with the segmentation head and then fine-tuned using limited annotations. This fine-tuning process aims to update the parameters of both the backbone network and segmentation head specifically for the semantic segmentation task in remote sensing scenarios.

where $H \times W$ is the input's spatial size, M denotes the total number of categories, $p_{nm} \in [0, 1]$ represents the predicted probability that nth pixel belongs to the mth category, $y_{nm} \in \{0, 1\}$ indicates whether the nth pixel belongs to the mth category.

IV. Experiments

In this section, we begin by presenting the datasets utilized in our experiments. Subsequently, we provide a comprehensive description of the experiment setup, including the evaluation metrics, implementation details, and comparison methods. Finally, we present and analyze the results obtained from the comparison and ablation experiments.

A. Dataset

To validate the effectiveness of our proposed method, three remote-sensing semantic segmentation datasets are selected to serve as the foundation for both pretraining and fine-tuning stages. These datasets are specifically selected to encompass diverse landscapes, challenging environmental conditions, and a wide range of land cover classes, enabling a comprehensive evaluation of our method's superiority. In Section IV-A, we provide detailed descriptions of each dataset, highlighting their key characteristics and properties.

1) ISPRS Potsdam Dataset: The International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam dataset[1] consists of 38 true orthophoto (TOP) tiles, each containing four bands (red, blue, green, and near-infrared), and has a spatial size of 6000 × 6000 pixels. The dataset is meticulously annotated with six categories for semantic segmentation: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. To facilitate self-supervised training and evaluation, we employ a sliding window strategy to crop the 38 tiles into 256 × 256 patches, resulting in a total of 21 888 images. Among them, 13 824 images are utilized for pretraining and fine-tuning our model, while the remaining images are reserved for assessing the performance of the fine-tuned model.

2) LoveDA Dataset: The LoveDA dataset [46] contains 5987 images with a spatial size of 1024 × 1024, and each image contains three spectral bands (red, blue, and green). The dataset is divided into two distinct parts: urban and rural, representing different environmental settings. It is meticulously annotated with seven categories for comprehensive semantic segmentation: background, buildings, roads, water, barren land, forests, and agriculture. To focus on the urban scenes, we specifically select the urban part of the LoveDA dataset. These urban images are further cropped into nonoverlapping patches of size 256 × 256 pixels. The training set consists of 18 496 images, which are employed for both pretraining and fine-tuning our model. In addition, a separate validation set comprising 10 832 images is utilized for testing and evaluating the performance of our trained model. It is worth noting that the labels for the testing set are not provided, necessitating the use of the validation set for performance assessment.

3) GID Dataset: The Gaofen image dataset [47] contains 150 tiles, each captured with three spectral bands: red, blue, and green. Notably, each tile encompasses a substantial spatial extent, measuring 7200 × 6800 pixels. The dataset provides annotations for five prominent land-cover classes, namely, built-up, farmland, forest, meadow, and water. Similarly, we employ the sliding window strategy to crop all the tiles into smaller patches of size 256 × 256 pixels. This procedure results in a considerable dataset size, with a total of 117 450 images. In accordance with the train-test split strategy [48], we allocate 78 300 images from the dataset for pretraining and fine-tuning the model. Subsequently, the remaining 39 150 images are designated for testing and evaluating the performance of our trained model.

B. Experiment Setup

In this section, we provide a comprehensive description of the evaluation metrics employed to assess the performance of various methods, elaborate on the implementation details of our proposed SSCCL framework, and present the comparison methods utilized in our experiments.

1) Evaluation Metrics: To ensure a rigorous quantitative evaluation of the methods employed in our experiments, we have selected three reliable and widely used evaluation metrics: overall accuracy (OA), F_1 score, and mean intersection-over-union (mIoU) of pixels.

The OA metric measures the percentage of correctly classified pixels across all classes, providing an overall indication of the algorithm's performance in terms of correctly predicting the correct class labels, which can be expressed as

$$\mathrm{OA} = \frac{(\mathrm{TP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})} \tag{12}$$

where TP, FP, TN, *and*FN stand for true positive, false positive, true negative, and false negative, respectively. These four indicators represent the situation when different pixels are predicted into different categories.

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/semanticlabeling

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The F_1 score combines precision and recall to evaluate the segmentation performance on a per-class basis. It considers both false positives and false negatives, making it an effective metric for assessing the algorithm's ability to capture class-specific details and discriminate between different land cover categories. The F_1 score is calculated using the following formula:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
$$\text{F}_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{13}$$

The mIoU metric evaluates the spatial consistency and overlaps between the predicted segmentation masks and the ground truth labels. It measures the average intersection-over-union for all classes, providing insights into the algorithm's ability to accurately delineate the boundaries and capture the extent of each land cover class. The mIoU metric can be expressed as follows:

$$\text{mIoU} = \frac{1}{\text{k}} \sum_{\text{i}=0}^{\text{k}} \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{14}$$

2) Implementation Details: In our experiment setup, we implemented the proposed SSCCL using the PyTorch framework [49]. The pretraining and fine-tuning process was conducted on four Nvidia RTX A100 GPUs equipped with 80 GB memory, allowing for efficient computation and optimization. For a fair comparison, all comparative methods apply the same experiment settings as our proposed SSCCL.

We select ResNet50 [43] as the backbone network, a widely recognized and well-established deep neural network architecture known for its strong feature extraction capabilities. Besides, we employ the DeepLabV3+ [50] to perform the semantic segmentation task.

During the pretraining phase, the images in the training set of each selected remote sensing semantic segmentation dataset are used to pretrain the backbone network from scratch. For an input image size of 256 × 256, we crop a random portion of each image and resize it to 224 × 224. To optimize the model's performance, we employ the Adam [51] optimizer with a weight decay of 1e-6. The initial learning rate is set to 0.01, and we utilize a cosine annealing schedule decay to gradually decrease the learning rate over time. We pretrain the backbone network for 400 epochs with a batch size of 256. A larger batch size in pretraining improves parallel computing efficiency, accelerates convergence, and enables the model to learn richer and more generalized feature representations through increased sample diversity. The model with the lowest loss during self-supervised pretraining is saved for subsequent semantic segmentation tasks.

After the pretraining phase, we apply a conventional fine-tuning strategy to further refine the DeepLabV3+ model, using annotations from the training set for each dataset to update the parameters of the backbone and decoder. We use the Adam optimizer with a weight decay of 1e-5 for 200 epochs, and the batch size is 16. Smaller batch sizes facilitate more accurate and precise parameter updates by providing a finer-grained representation of the training data in each batch. The initial learning rate is set to 1e-4, and we employ the "Poly" decay strategy [52] to dynamically adjust the learning rate. We utilize limited annotations, typically 1% of the total amount of self-supervised training data with annotations, to fine-tune the parameters of the DeepLabV3+ model. The image size used during fine-tuning is set to 256 × 256 pixels.

3) Comparison Methods: To assess the effectiveness of the proposed method, we consider random initialization and conventional ImageNet pretraining strategy as the baseline. Besides, we compare our proposed SSCCL with the other nine CL methods, which are summarized as follows.

1) *MoCo [28]:* MoCo leverages momentum updates and query caching for efficient negative sampling. By utilizing a dynamically updated cache of historical feature vectors, MoCo is able to sample more high-quality negatives in the generated samples.
2) *MoCo v2 [42]:* MoCo v2 is an upgraded version of MoCo, which is optimized in terms of sampling strategy, extended dictionary, and data augmentation, and proposes a more efficient backpropagation method to further improve the performance of self-supervised learning.
3) *SimCLR [17]:* SimCLR achieves efficient image representation learning by constructing a large number of positive and negative sample pairs and designing a novel contrastive loss function, providing a novel paradigm for self-supervised learning.
4) *SimSiam [18]:* SimSiam proposes a simple Siamese network to learn representations without the need for negative pairs, large batches, and momentum encoding.
5) *BYOL [29]:* BYOL uses two neural networks, called the online network and the target network, that interact and learn from each other. In addition, BYOL does not need to use negative pairs, it iteratively guides the output of the network to serve as the target of the enhanced representation.
6) *BarlowTwins [53]:* The optimization goal of BarlowTwins is to make the correlation matrix of features under different perspectives close to the identity matrix, so that features of different dimensions can represent different information as much as possible, thereby improving the representation ability of features.
7) *DenseCL [54]:* DenseCL is a dense CL algorithm that implements self-supervised learning by optimizing the pairwise contrastive loss between two views of the input image at the pixel level.
8) *GLCNet [35]:* GLCNet is a self-supervised CL framework for remote sensing semantic segmentation task, aiming to balance the global and local feature learning.
9) *IndexNet [55]:* The index contrast mechanism is introduced in IndexNet to take into account the multiobject characteristics and the complex interactions between objects, so as to maintain the spatial transformation invariance in the process of learning feature representation.

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TABLE I

COMPARISON RESULTS WITH RANDOM INITIALIZATION, IMAGENET PRETRAINING, AND OTHER CL METHODS ON THE POTSDAM DATASET

Methods	OA (%)	F_1 (%)	mIoU (%)
Random Init	66.36	56.13	41.77
ImageNet	75.31	65.69	52.71
MoCo	71.83	62.58	49.00
MoCo v2	74.80	65.45	52.49
SimCLR	75.60	65.93	52.89
SimSiam	72.29	62.48	48.59
BYOL	74.45	65.31	51.85
BarlowTwins	71.65	62.72	48.94
DenseCL	73.40	64.04	50.48
GLCNet	75.29	66.51	52.03
IndexNet	75.66	66.87	52.91
SSCCL (ours)	**76.10**	**67.14**	**53.40**

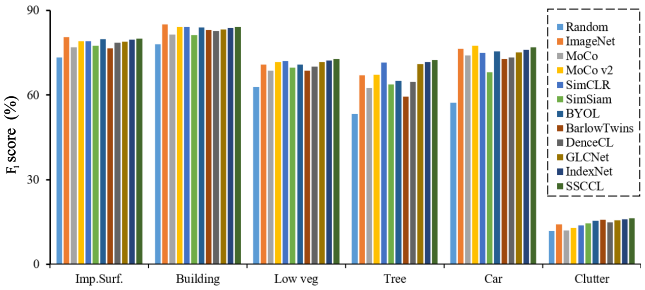


Fig. 5. F_1 score for each category predicted by fine-tuned segmentation model with different CL methods on the Potsdam dataset.

C. Experiment Results

1) Potsdam Dataset: We evaluated the segmentation performance of the proposed SSCCL after fine-tuning with limited annotations on the ISPRS Potsdam dataset. The comparison results are presented in Table I, where we compared our method with the random initialization, ImageNet pretraining, and other contrastive CL methods. When applied to the semantic segmentation task fine-tuned with 1% annotations, our proposed SSCCL outperforms random initialization and ImageNet pretraining. It is worth noting that ImageNet pretraining relies on supervised training using millions of labeled samples, which greatly surpasses the amount of data used for self-supervised training in the Potsdam dataset. This highlights the challenge of transferring knowledge learned from natural images to remote sensing scenarios due to domain differences. Our proposed SSCCL framework, which leverages a large amount of unlabeled remote sensing data, proves to be a more effective way to learn general representations from remote sensing images. The superior performance of SSCCL demonstrates its capability to capture meaningful features and improve the segmentation accuracy on the Potsdam dataset.

Furthermore, as observed from Table I, our proposed SSCCL demonstrates the highest segmentation performance among the compared CL methods. This indicates that SSCCL enables the backbone network to learn more comprehensive feature representations, which greatly benefits the semantic segmentation task. In addition, we calculate the F_1 score for each category, as shown in Fig. 5. Our SSCCL consistently achieves the best performance across almost all categories, which demonstrates the robustness and generalization capability of SSCCL. Furthermore, Fig. 6 visually illustrates the superior segmentation performance of our method compared to other CL methods.

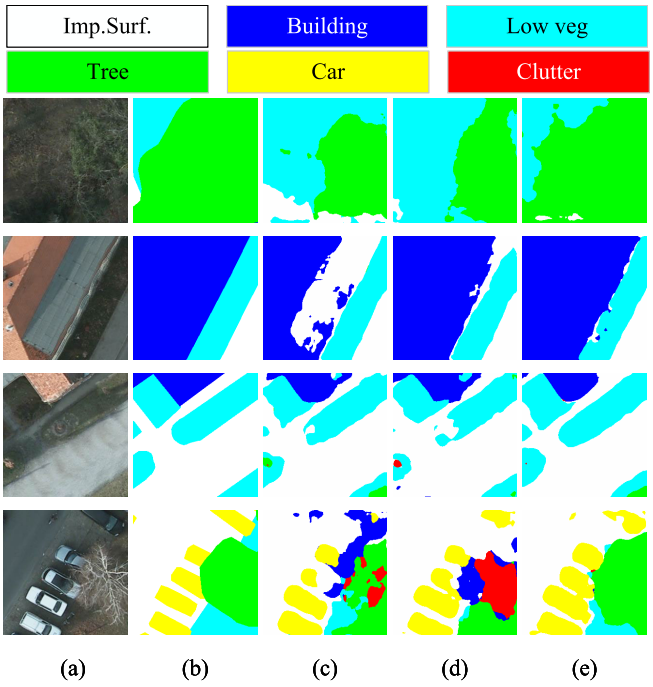


Fig. 6. Visualization results of our method and compared CL methods on Potsdam test dataset. (a) Images. (b) Ground truth. (c) MoCo. (d) SimSiam. (e) SSCCL.

TABLE II

COMPARISON RESULTS WITH RANDOM INITIALIZATION, IMAGENET PRETRAINING, AND OTHER CL METHODS ON THE LOVEDA DATASET

Methods	OA (%)	F_1 (%)	mIoU (%)
Random Init	40.56	42.86	31.06
ImageNet	47.92	52.03	38.84
MoCo	40.98	45.15	32.98
MoCo v2	47.30	52.63	39.83
SimCLR	49.48	55.18	41.60
SimSiam	44.14	47.71	35.15
BYOL	48.80	53.69	40.34
BarlowTwins	49.50	55.44	41.76
DenseCL	48.71	53.85	39.29
GLCNet	49.26	55.17	41.62
IndexNet	50.14	55.78	42.24
SSCCL (ours)	**50.43**	**56.05**	**42.61**

2) LoveDA Dataset: We also conducted comparative experiments on the LoveDA dataset to evaluate the effectiveness of our proposed method, and the results are presented in Table II. Using the same experiment settings, SSCCL outperforms all other CL methods. Furthermore, our method surpasses random initialization and ImageNet pretraining by 11.55% and 3.77% in terms of mIoU, respectively, highlighting the importance of leveraging self-supervised CL techniques specifically tailored for remote sensing applications. In addition, Fig. 7 showcases several samples from the LoveDA urban test set, illustrating the improved segmentation results achieved by SSCCL.

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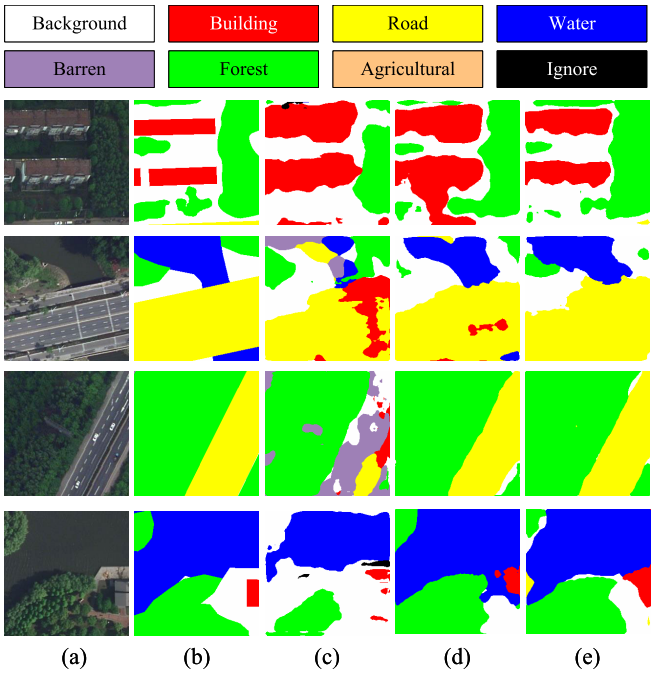


Fig. 7. Visualization results of our method and compared CL methods on LoveDA urban test dataset. (a) Images. (b) Ground truth. (c) MoCo v2. (d) BarlowTwins. (e) SSCCL.

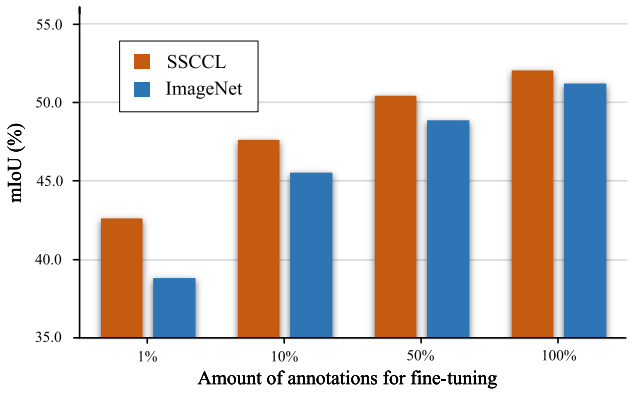


Fig. 8. Segmentation performance after fine-tuning with different amounts of annotations on the LoveDA dataset.

TABLE III

COMPARISON RESULTS WITH RANDOM INITIALIZATION, IMAGENET PRETRAINING, AND OTHER CL METHODS ON THE GID DATASET

Methods	OA (%)	F_1 (%)	mIoU (%)
Random Init	66.43	54.34	41.25
ImageNet	75.99	64.10	51.41
MoCo	70.67	61.34	47.55
MoCo v2	73.14	64.17	50.51
SimCLR	74.45	64.72	51.71
SimSiam	68.44	69.60	46.46
BYOL	75.11	65.35	52.21
BarlowTwins	67.91	57.62	44.40
DenseCL	77.22	65.85	53.41
GLCNet	77.04	66.26	53.68
IndexNet	77.14	66.92	54.11
SSCCL (ours)	**77.81**	**67.64**	**54.84**

These visual examples further reinforce the effectiveness and robustness of our proposed method in capturing fine-grained details and accurately delineating urban land cover classes.

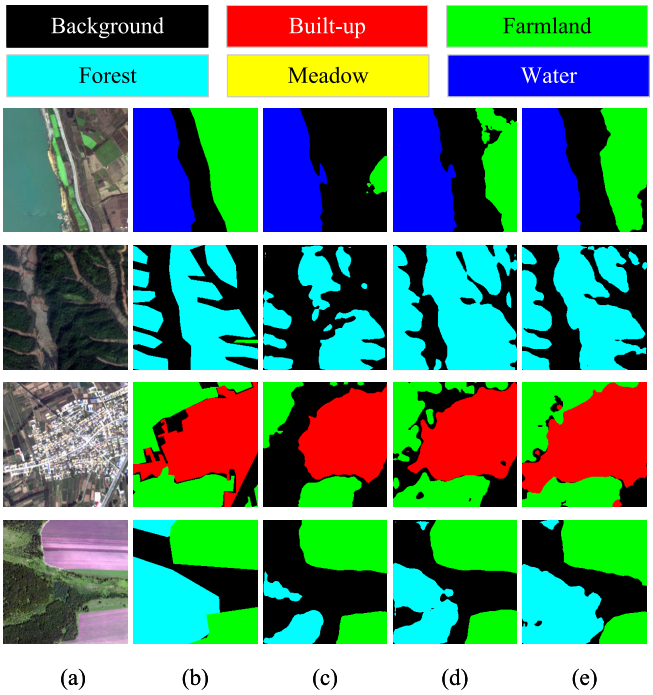


Fig. 9. Visualization results of our method and compared CL methods on GID test dataset. (a) Images. (b) Ground truth. (c) BYOL. (d) DenseCL. (e) SSCCL.

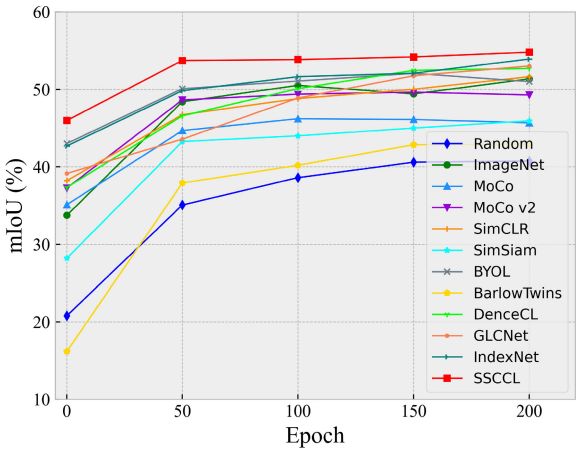


Fig. 10. Segmentation performance at different epochs in the fine-tuning phase on the GID dataset.

Since obtaining pixel-level annotations in remote sensing scenes is expensive and challenging, we investigate the performance of our proposed SSCCL under varying amounts of annotations. To this end, we conduct experiments on the LoveDA dataset by randomly selecting 1%, 10%, 50%, and 100% of annotated data for fine-tuning. We compare our method with ImageNet pretraining to assess the impact of annotation scarcity. The comparison results are shown in Fig. 8. From the results, we observe that SSCCL consistently outperforms ImageNet pretraining across different amounts of annotated data. Particularly, the superiority of our proposed method becomes more apparent as the availability of annotated data decreases. This highlights the suitability of SSCCL for practical remote sensing scenarios, where the availability of large-scale annotated data is limited. By leveraging

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TABLE IV
EXPERIMENT RESULTS OF ABLATION STUDIES EXPLORING THE EFFECTIVENESS OF THE MODULES OF OUR PROPOSED SSCCL ON POTSDAM, LOVEDA, AND GID DATASETS

Method	Potsdam			LoveDA			GID		
	OA	F1	mIoU	OA	F1	mIoU	OA	F1	mIoU
Ours	**76.10**	**67.14**	**53.40**	**50.43**	**56.05**	**42.61**	**77.81**	**67.64**	**54.84**
Ours_w/o_Instance_Branch	75.62	66.03	52.95	49.21	53.87	39.16	75.26	65.47	52.40
Ours_w/o_Consistency_Branch	75.26	65.38	51.61	49.67	54.54	40.11	75.03	64.86	51.77

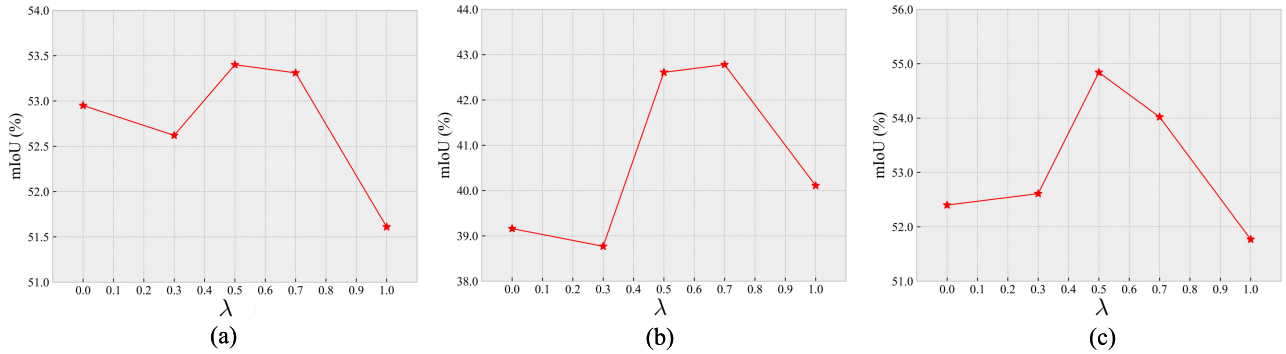


Fig. 11. Ablation study of the value of hyperparameter λ on (a) Potsdam, (b) LoveDA, and (c) GID datasets.

self-supervised CL learning on remote sensing data, our approach overcomes the limitations of relying solely on ImageNet pretraining, which is based on natural images. This underscores the practical significance of our method for remote sensing applications that lack a substantial amount of labeled data.

3) GID Dataset: The comparison results for the GID test set are summarized in Table III. It is evident that all CL methods contribute to performance improvements in the backbone network. Notably, our proposed method achieves the most substantial enhancements, surpassing random initialization. Specifically, our method demonstrates a remarkable increase of 13.30% and 13.59% in the mIoU and F_1 metrics, respectively. Besides, we present representative visual results in Fig. 9. The visual comparison reveals that our method excels in category prediction accuracy and object edge delineation, outperforming BYOL and DenseCL.

Furthermore, our proposed method exhibits superior performance during the initial stages of the pretraining phase compared to other CL methods, as shown in Fig. 10. This observation suggests that our approach facilitates the backbone network in acquiring and capturing essential features and representations. In addition, our SSCCL has reached the optimal level at about 50 epochs, and the segmentation performance has basically remained unchanged during the subsequent fine-tuning process. It demonstrates that our method helps the segmentation network to fit and converge faster in the fine-tuning stage, which is very beneficial for resource-constrained real-world remote sensing scenarios.

D. Ablation Study

In this section, we perform ablation studies on the Potsdam, LoveDA, and GID datasets to validate the effectiveness of the modules in our SSCCL framework and assess the impact of different values of weight hyperparameter λ. We adopt the same experiment settings as in the previous experiments, which involve a pretraining phase of 400 epochs followed by a fine-tuning phase of 200 epochs.

1) Effectiveness of the Modules of SSCCL: The experiment results are presented in Table IV, where the entries labeled as "Ours_w/o_Instance_Branch" and "Ours_w/o_Consistency_Branch" indicate the scenarios where the instance branch and consistency branch are completely removed from our proposed SSCCL method, respectively. From the table, it is evident that the complete SSCCL achieves the highest performance across all three datasets, indicating the positive impact of each module on segmentation performance.

Moreover, we observe that when the consistency branch is not applied, our method exhibits the lowest performance on both the Potsdam and GID datasets. This finding highlights the importance of spatial-semantic-level feature representation learning for accurate semantic segmentation of remote sensing images. The inclusion of the consistency branch enables the model to capture and leverage spatial and semantic relationships effectively, resulting in improved segmentation results. Whereas on the LoveDA dataset, the performance of "Ours_w/o_Instance_Branch" surpasses that of "Ours_w/o_Consistency_Branch." This observation suggests that the global representation provided by the instance branch plays a more crucial role in scenarios where categories are prone to confusion. By emphasizing the overall scene information, the instance branch helps alleviate the confusion between similar categories, leading to enhanced segmentation accuracy.

2) Ablation Study of Weight λ: The final loss $\mathcal{L}$ for self-supervised pretraining comprises two components: the instance-level similarity loss $\mathcal{L}_I$ and the spatial-semantic-level similarity loss $\mathcal{L}_S$. To balance their contributions,

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a hyperparameter λ is introduced. In our article, we set λ to 0.5 as a default value, considering the significance of both the instance branch and the consistency branch. However, the optimal value of λ may vary depending on the specific context.

In this section, we aim to investigate the impact of different λ values on segmentation performance. As shown in Fig. 11, when λ is set to either 0.0 or 1.0, the model tends to exhibit the poorest segmentation performance. This finding further reinforces the importance of both instance-level representation learning and spatial-semantic-level representation learning for achieving accurate semantic segmentation. Moreover, we observe that for the Potsdam and GID datasets, setting λ to 0.5 yields optimal results, indicating the significance of emphasizing spatial-semantic-level similarity. Conversely, for the GID dataset, the best performance is achieved with $\lambda = 0.7$. Therefore, the value of λ is fixed at 0.5 for generalizability.

V. Conclusion

In this work, we introduce a novel SSCCL framework, which specifically addresses the limitations of existing CL methods in remote sensing semantic segmentation. By integrating spatial and semantic consistency learning, our framework effectively captures the complex spatial relationships and semantic coherence required for accurate segmentation. The experiment results validate the effectiveness of our approach in remote sensing scenarios, highlighting its potential for enhancing segmentation performance with limited annotated data. Moving forward, the proposed SSCCL framework opens up new avenues for further research and advancements in remote sensing semantic segmentation, contributing to improved scene understanding and decision-making in various application domains.

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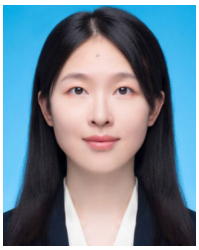
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